CO3-AT1-Assignment

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QUESTIONS

Q4 Online Food Delivery – Multiple Threads and Execution Order

10 marks

An online food-delivery application uses three threads: Order Processing, Payment Processing, and Delivery Assignment.

Develop a Java program in which these activities are represented by separate threads. Analyze the problem if all three threads are started independently.

Modify the program so that:

- Order processing completes before payment processing.
- Payment processing completes before delivery assignment.
- The program uses appropriate Java thread-management techniques.
- The output clearly shows the required execution order.

Explain the difference between simply starting all threads and explicitly controlling their execution order.

Sample Test Cases:

Input / Scenario Expected Result Analysis Focus

Order=Pizza; Payment=UPI; Delivery=Agent A Order → Payment → Delivery Normal execution

10 marks

5/5 answered

Order=Burger; Payment=Card; Delivery=Agent B Order → Payment → Delivery Concurrency / design Payment=Failed Delivery assignment should not proceed Exception / boundary

Your Code

```
1 import java.util.Scanner;
2
3 class OnlineFoodDelivery{
4     static boolean paymentSuccess = false;
5
6     static class OrderThread extends Thread{
7         String order;
8
9         OrderThread(String order){
10            this.order=order;
11        }
12        public void run(){
13            System.out.println("Order Processing :"
14        }
15    }
16    static class PaymentThread extends Thread{
17        String payment;
18
19        PaymentThread(String payment){
20            this.payment = payment;
21        }
22        public void run(){
23            if(payment.equalsIgnoreCase("failed")){
24                paymentSuccess=false;
25                System.out.println("Payment failed"
26            }else{
27                paymentSuccess = true;
```

Input

```
pizza UPI AgentA
```

Output

```
Order Processing :pizza
Payment successful:UPI
Delivery assinged :
AgentA
```

```
28                System.out.println("Payment success
29            }
30        }
31    }
32    static class DeliveryThread extends Thread {
33        String delivery;
34
35        DeliveryThread(String delivery){
36            this.delivery = delivery;
37        }
38        public void run(){
39            if(paymentSuccess){
40                System.out.println("Delivery assing
41            }else{
42                System.out.println("Delivey assigne
43            }
44        }
45    }
46
47    public static void main(String[]args)throws
48        Exception{
49
50        Scanner sc = new Scanner(System.in);
51
52        String order = sc.next();
53        String payment = sc.next();
54        String delivery = sc.next();
55
56        OrderThread orderThread = new OrderThread(o
57        PaymentThread paymentThread = new PaymentTh
58        DeliveryThread deliveryThread = new Deliver
59
60        orderThread.start();
```

```
61          orderThread.join();
62
63          paymentThread.start();
64          paymentThread.join();
65
66          deliveryThread.start();
67          deliveryThread.join();
68      }
69  }
```

Submitted